

Monitoring Inter-Drone Service Interference for Resilient Operations

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Abstract—We propose a novel service-based framework for drone service resilience. Our framework monitors inter-drone interference that may lead to drone service failure. We present a novel drone service interference taxonomy to formally identify different interference types in a skyway network. We then propose a heuristic-based approach that leverages spatio-temporal proximity analysis to detect the occurrence of inter-drone interference. In addition, we present an interference severity assessment to quantify their impact on drone services' efficiency. We conduct a set of experiments using real-world datasets to evaluate the effectiveness and efficiency of our proposed approach. The results indicate that the proposed heuristic-based approach detects the occurrence of inter-drone interferences with an accuracy of 95%. In addition, the proposed method is $\approx 70\%$ more efficient than the baseline exhaustive approach and $\approx 48\%$ faster than the K-means approach.

Index Terms—Drone delivery, drone service, skyway network, inter-drone interference, unmanned traffic management.

I. INTRODUCTION

THE demand for ubiquitous deliveries from warehouses to customers' doorsteps has increased due to the expansion of urban areas. Traditional ground transportation faces challenges in ensuring seamless delivery operations for e-commerce companies due to growing road congestion. Unmanned Aerial Vehicles (UAVs) or drones have the potential to deliver packages more efficiently than traditional transportation [1]. These drones efficiently maneuver over various terrains to navigate their route in the airspace to ensure ubiquitous and prompt deliveries [2].

Existing work on drone-based delivery services leverages the service paradigm to model Drone-as-a-Service (DaaS). A *drone service* is defined in terms of its functional and non-functional attributes [3]. The functional attribute of a drone service refers to the transportation of a package between two nodes [4]. The non-functional attributes are the Quality of Service (QoS) parameters such as energy consumption, cost, and delivery time. Drones associated with these services operate within a *skyway network* [4]. A skyway network consists of an interconnected

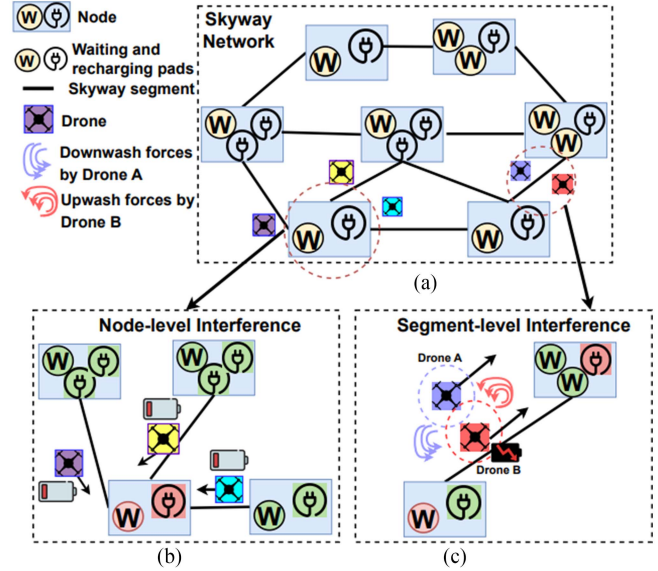


Fig. 1. (a) Skyway network (b) and (c) Interferences in a skyway network.

set of *nodes*, serving as the endpoints of *skyway segments* (Fig. 1(a)). The skyway segment is a Line of Sight (LoS) path between the nodes. The nodes are the building rooftops that may serve as both charging stations and delivery destinations for drones. The skyway network restricts drone entry in specific areas, defined based on airspace safety regulations.¹ These areas include No-Fly-Zones (NFZs), and Temporary-No-Fly-Zones (TNFZs).

Recent research focuses on drone service composition to ensure on-time package delivery [4]. The proposed approaches optimize the composition of *drone services* based on specific QoS requirements. These *drone services* refer to an optimal set of drone-operated segments. However, a key assumption in existing solutions is that drones operate *smoothly* without interference from other drones. This assumption is unrealistic, as the expected growth in drone traffic will result in drones flying in close proximity [5]. Drones traveling closely may result in drones impacting each other [6]. For example, aerodynamic forces between closely flying drones may increase their energy consumption. This interference may cause drones' longer charging times, impacting the expected QoS requirements, e.g., delivery time.

¹ casa.gov.au/drones/drone-rules/emergencies-and-public-spaces

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Airbus's recent study forecasts 16,000 drone flights per hour in Paris by 2035, within an area of 105.4 km^2 [7]. This density translates to 79 simultaneous flights per square kilometer, with each flight lasting about 30 minutes. As large drones (e.g., DJI Flycart 30, Amazon MK30, EHangFalcon) carrying payloads of up to 10 kg are expected to operate within delimited airspace, such high traffic density will inevitably result in drones operating in close-proximity within shared skyway segments [5], [8]. Furthermore, this likelihood increases as each provider plans routes independently and selfishly, often prioritizing the shortest paths for efficiency. This may result in multiple drones traversing the same segment in close proximity, potentially impacting drone performance due to aerodynamic turbulence generated by nearby drones [9]. For instance, a drone may consume more battery power than initially anticipated to counteract turbulence. This increase in battery power consumption occurs as the drone's motors draw extra power to rotate the propellers faster to maintain stability within the turbulent region. This may cause drones' longer charging times or the risk of a mid-air emergency landing due to excessive battery depletion, impacting the expected QoS requirements (e.g., delivery time, safety). Therefore, this significant increase in drone traffic demands that the DaaS skyway infrastructure adapts to ensure the *smooth* operation of drones.

An end-to-end Service Oriented Architecture (SOA) is required to orchestrate *smooth* drone operations in the skyway network. We identify three main components of a drone-based SOA: *service-provider*, *service-consumer*, and *super-provider*. A *service-provider*² is the drone owner who offers delivery services. *Service-consumers* are the end customers who invoke drone services from service providers based on their delivery requirements. A *super-provider* manages the skyway network to ensure the *smooth* operation of drones.

Smoothness refers to ensuring the *resilient* operation of drone services. A *resilient* drone service adapts to the *failures* occurring in the skyway network. We define *failure* as an event that impacts the *safety* and *efficiency* of drone services. These *failures* may result in *collisions* or *near collisions* (a.k.a *near misses*). A *collision* occurs when drones come in direct contact, compromising *safety*. For instance, physical damage may occur when two drones traveling in opposite directions converge at the same altitude in the same skyway segment. In contrast, *near-collision* affects the *efficiency* of drones when they operate in close *spatio-temporal proximity*. For instance, the aerodynamic interference between nearby drones may increase their energy consumption [6]. As a result, drones may deviate from their expected trajectory. This deviation may cause a risk of damage or a later arrival than their scheduled delivery time. Hence, addressing *failures* that may impact the *safety* or *efficiency* of drones is crucial to ensure the *smooth* operation of drones. Existing research focuses on the detection of *collisions* that impact drones *safety* [10]. In this work, we focus on the *failures* that may impact the *efficiency* of drones.

Failures occur due to inter-drone **interference**. We define *interference* as any *spatio-temporal* event that hinders the *efficient* operation of drones. In the context of service-based

drone deliveries, we propose a novel drone service interference taxonomy to classify various types of *interference* that may occur between drone services. Interference may occur at the **node-level** or **skyway segment-level** as shown in Fig. 1. A **node-level** interference is defined as any event that delays the *provisioning of drones' recharging requirements* at the nodes. For example, node-level interference may occur when there are insufficient recharging pads to accommodate multiple drones' access requests (Fig. 1(c)). A **segment-level** interference refers to an event that impacts the expected segment-level requirements of drones (e.g., energy consumption). This interference may occur due to a *spatio-temporal* overlap in the trajectories of drones operating in the same segment. As a result, the expected energy consumption of drones is influenced either *positively* or *negatively*. For example, a *positive segment-level interference* may occur when the drone consumes less battery than anticipated due to upwash forces generated by a drone flying below it (Fig. 1(c)). In contrast, a *negative segment-level interference* may occur when a drone consumes more battery than initially expected due to the downwash forces from another drone situated above it (Fig. 1(c)).

There may be instances where *segment-level interference* may lead to *node-level interference*. For instance, increased energy consumption due to segment-level interferences may cause multiple drones to arrive at the subsequent nodes to recharge for a longer duration than initially planned. This may result in a congestion buildup at the node, causing the drones to wait for their turn to occupy the recharging pads. In this paper, we focus on the proactive *detection of negative segment-level inter-drone interference*. This detection serves as a foundation for effective interference *resolution* strategies by mitigating the propagation of interferences to the *node-level* for *resilient* drone services' operation. External factors such as GPS jamming, spoofing, thunderstorm sounds, and birds may disrupt the data reception and transmission between drones. As a result, it may threaten the accurate detection of inter-drone interferences. However, addressing these factors is out of the scope of this paper. The prior research studies have addressed the detection and resolution of these types of interference [11]. A key aspect of resolving *segment-level interferences* is assessing the severity of interference on the expected QoS requirements of drone services. Some interferences are critical and require immediate resolution, while others may be negligible. For instance, a drone may consume more battery due to a long duration of spatio-temporal overlap with the peer drone. As a result, it may arrive late at the destination due to an increase in recharging time at the intermediate node. However, the impact of some interference events may be negligible. For instance, the expected battery consumption of drones may get negligibly impacted if drones operate in close proximity for a few seconds. In this case, drones may arrive at their scheduled destination with a delay of a few minutes. Therefore, it is crucial to assess the severity level to devise effective interference resolution strategies.

Detecting and assessing the severity of inter-drone interferences is challenging due to the *dynamic spatio-temporal requirements* of multiple drone providers using the skyway network. For example, each provider may operate drones at different

²From now onwards, we will use "drone providers" and "service-providers" interchangeably.

altitudes and times across multiple segments to meet specific QoS requirements. Additionally, the impact of interference may vary based on the *spatial proximity* and *spatial configuration* of drones operating in the same segments. For instance, drones with a vertical separation of 30 cm may experience significantly greater interference than drones 80 cm apart. Another example, a drone operating in a downwash region may be significantly more affected compared to a drone operating in an upwash region. To address the aforementioned challenges we propose a novel *segment-level interference detection framework* to proactively *detect* and *assess the severity level* of inter-drone interferences in a skyway segment. The main contributions of this paper are as follows:

- **Novel Inter-drone Interference Detection Framework:** We propose a service-based framework to monitor and assess inter-drone interference in skyway networks. It detects interferences by analyzing spatio-temporal dynamics in 3D space and evaluates their severity for effective mitigation.
- **Comprehensive Interference Taxonomy:** Our taxonomy categorizes interferences into node-level (e.g., recharging pad congestion) and segment-level (e.g., aerodynamic interactions). This classification helps devise tailored strategies for addressing interferences based on their type and context.
- **Segment-level Interference Detection Algorithm:** We developed a heuristic-based algorithm to detect interference events in 3D space efficiently. It uses real data to evaluate spatio-temporal proximity and performs faster than baseline methods.
- **Segment-level Interference Severity Assessment Method:** We introduce a severity assessment using the *Percentage Impact Score (PIS)* metric. This method quantifies the impact of interferences on energy consumption and delivery times. This enables prioritized and effective mitigation measures to address critical interferences.

The current approach for inter-drone interference detection assumes that the super-provider is aware of the drone trajectories. This assumption would be relaxed in future work as *uncertainties in weather conditions* and the *trustworthiness of drone providers* may compromise the completeness of super-providers' knowledge and, consequently, the accuracy of pre-flight inter-drone interference detection. For instance, a drone provider may intentionally extend their drone's use of a recharging pad during peak demand. Similarly, drones may deviate from their planned path due to unfavorable weather conditions during flight operations. Future work will address these factors to allow dynamic interference detection during real-time drone operations.

II. MOTIVATING SCENARIO

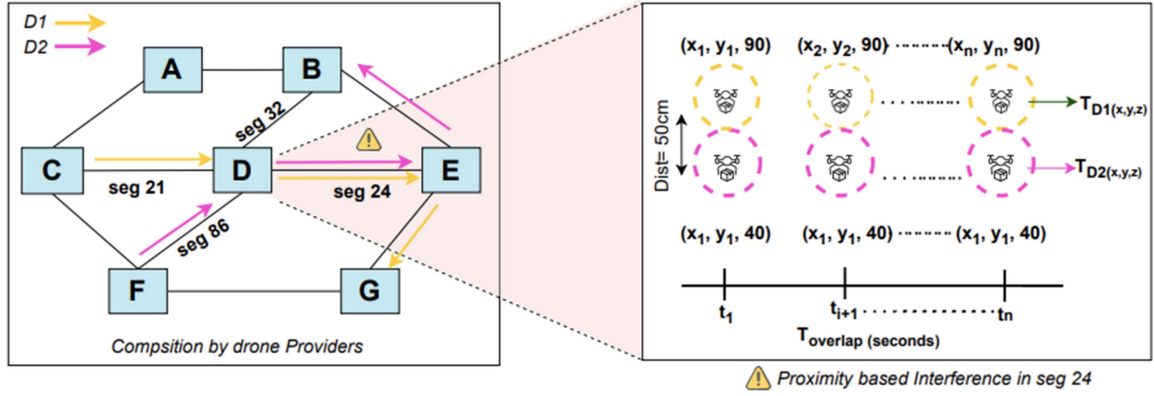
Let us consider a scenario where multiple *drone providers* operate single drones in a skyway network. Each drone provider composes an optimal set of *drone services* to deliver packages from the source node to the destination node [4]. We assume

that drone providers comply with the regulatory practices established by the Civil Aviation Safety Authority (CASA). In this regard, they submit detailed flight plans to the super-provider to ensure safe airspace utilization [12]. Therefore, the super-provider is aware of the drone trajectories and recharging duration at the nodes. The drone services' expected QoS requirements must be fulfilled to ensure the package delivery on the scheduled time. However, the expectations of drone providers may not be fulfilled due to inter-drone *interferences* occurring in segments. As a result, drones may arrive later than initially anticipated by the drone providers. We discuss a scenario of inter-drone interference that impacts the *efficient* operation of drones resulting in a late package delivery.

Consider a scenario where two drone providers, *DP1* and *DP2*, own drones *D1* and *D2*, respectively. *DP1* is required to deliver a package to destination node *B* in 54 minutes, while *DP2* must deliver a package to destination node *G* in 35 minutes. However, the interference in *seg 24* causes the actual delivery times to be 82 minutes and 61 minutes, respectively (Fig. 2). The actual delivery times due to the occurrence of interferences are computed using the (22) given in Section V-B. As shown in Fig. 2, both drones operate in close spatial proximity with a vertical separation of 50cm for an overlapping time interval of 300 seconds (i.e. 5 minutes). As a result, the actual battery consumption of drone *D1* due to the impact of interference from the peer drone *D2* is $\approx 67\%$ ((19)). Consequently, drone *D1* requires to spend ≈ 28 minutes ((23)) in actual at the intermediate node *E* to recharge. Similarly, drone *D2* requires to spend ≈ 26 minutes at the intermediate *E* node to regain the drained battery. Both these drones consume more battery than initially anticipated due to which they spend more time recharging at the intermediate node. Consequently, they arrive late at their respective destination nodes. However, some drones may experience a negligible impact due to interference from peer drones. For example, the impact on the battery consumption of the drone would be negligible if the overlap is for a few seconds. Therefore, a framework is required to proactively *detect* and *assess* the severity of interferences. This framework would allow the super-provider to manage drone traffic by devising effective interference *resolution* or *prevention* measures. For instance, in the above scenario, drone *D2* may be re-directed to *seg 32* to ensure the fulfillment of their expected delivery time requirement (i.e., 35 minutes). Interference *prevention* and *resolution* strategies are out of the scope of this paper. In this paper, we focus on the *detection* and *severity assessment* of these interferences. We aim to answer the following question: *Given n drones with their segment-level requirements, how do we accurately detect and assess the intensity of segment-level interference in a skyway network?*

III. RELATED WORK

We identify two research areas that are relevant to our work. (1) Service-based drone deliveries (2) Conflict detection in UAV traffic management.



Drone ID	Expected Battery Consumption	Expected Delivery Time	Spatio-temporal overlap (S_k, T_{overlap})	Peer Drone ID	Actual Battery Consumption	Increase in recharging time	Actual Delivery Time	Service status
D1	34%	54 mins	(seg 24, 5mins)	D2	~67%	28 mins	82 mins	Late
D2	40%	35 mins	(seg 24, 5mins)	D1	~73%	26 mins	61 mins	Late

Fig. 2. Segment-level interference in a skyway network.

A. Service-Based Drone Deliveries

Logistic companies are globally transforming their business models to provide drone-based delivery services [13]. There has been a growing interest in using the service paradigm for composing the right set of drone services to ensure the package delivery from source to destination node based on QoS parameters. The recent work in service-based drone deliveries proposed an algorithm for optimal drone service selection and composition [14]. However, the drone service environment is highly dynamic due to uncertain weather conditions. This uncertainty may lead to the unavailability of recharging pads at nodes, thus leading to a delayed package delivery. A resilient DaaS composition framework is proposed considering the uncertainty in the availability of recharging pads due to weather conditions [15]. However, the proposed framework does not consider the uncertainty of wind conditions in skyway segments. A drone service composition approach is proposed considering the probability of wind speed and direction in each skyway segment using the Probabilistic Forward Search (PFS) algorithm [16] as the uncertainty in wind speed and direction impacts the optimal composition of drone services.

Recent work also focuses on using a *swarm of drones* for package deliveries. A *swarm-based drone* for package delivery refers to a set of drones that operate together from a source to a destination node within time and spatial constraints [17]. A swarm of drones leverages faster and more efficient package delivery by alleviating the limitations (e.g., limited payload capacity) associated with single drone-based package delivery [6]. A novel swarm-based drones-as-a-service (SDaaS) model is proposed that leverages cooperative behavior among drones to reduce the waiting time associated with the sequential charging at nodes [18]. However, the proposed approach does not consider the dynamic wind conditions inherent to a drone service environment. A study is conducted for different types of drone

formations and their impact on energy consumption in varying wind conditions [6]. The proposed algorithm informs optimal drone formation, ensuring the least energy consumption in the segments. Unlike single-drone-based package delivery, swarm-based package delivery requires multiple drones to recharge at intermediate nodes. The drones may recharge fully or partially at intermediate nodes depending on the drone services' requirements. However, the skyway network is a constrained space with limited recharging pads. An in-flight energy-sharing framework is proposed for swarm-based drone services considering delivery time requirements [19]. The proposed framework enables the optimal set of support drones to share energy among drones. As a result, the energy sharing minimizes the probability of drone swarm waiting or stopping at intermediate nodes for recharging. The experimental results show that the priority-based energy sharing methods perform better in delivering packages successfully to the destinations.

These single-drone and swarm-based drone delivery services compete for airspace usage to fulfill their expected requirements in terms of service composition. This increasing competition may lead to interference between drones' operations, thus impacting their expected QoS requirements. *To the best of our knowledge, no existing studies on service-based drone deliveries have examined the impact of inter-drone interferences on drone service composition.* Hence, this work is the first attempt to monitor the usage of skyway segments to *detect* and *assess* the impact of *interferences* on drone services' *efficiency*.

B. Conflict Detection in UAV Traffic Management

FAA and NASA introduced the Unmanned Traffic Management (UTM) framework for maintaining safe separation between drones, given the increasing volume of drone traffic [20]. This framework manages drone flights in compliance with air traffic regulations. Research in UAV traffic management mainly

focuses on maintaining the conflict-free operation of drones in shared airspace. The current UTM models define *conflict* as any event that results in *collision* due to loss of safe separation between drones [10]. Several UTM models are proposed to mitigate UAV conflicts through *conflict detection* and *resolution*. *Conflict detection* identifies possible loss of safe separation between UAVs, leading to safety hazards such as collisions. In contrast, *conflict resolution* refers to resolving conflicts to ensure safe separation between UAVs.

Recent literature presents deterministic approaches for conflict detection among UAVs [21], [22], [10]. The work in [21] identifies the conflicting paths between flight pairs. In this respect, they determine the conflicting regions as overlapping paths between two UAVs. A state-based conflict detection strategy is proposed based on the extrapolation of drones' trajectories [22]. These extrapolated trajectories indicate the breach in safe separation between drones. A dynamic conflict detection framework is proposed based on trajectory extrapolation through historical observation data of UAVs [23]. The proposed framework leverages the Interacting Multi-Model (IMM) for trajectory tracking. The proposed approach uses the Gilbert–Johnson–Keerthi (GJK) algorithm to overcome the issue of low computational efficiency in traditional conflict detection approaches. A collision detection approach is proposed based on the collision cone concept. The proposed approach identifies a collision event by analyzing the position, velocity, and heading of both the obstacle and the UAV.

Existing drone-based frameworks focus on detecting potential collisions between drones. To the best of our knowledge, there is no work on detecting interferences that impact the *efficient* operation of drones in the skyway network. This work is the first attempt to develop an inter-drone interference detection framework in the context of service-based drone deliveries. This framework will serve as a pre-requisite to ensure *resilient* operation of drone services.

IV. PRELIMINARIES

In this section, we formally define segment-level requirements associated with a drone provider's request. We then present the notion of *skyway segment state* which reflects the dynamic condition of drone traffic operating in the segment. We also present the interference taxonomy in the context of service-based drone deliveries.

Definition 1 (Drone Provider Request (DPR)): A drone provider request consists of a set of optimal requirements associated with a drone service. The fulfillment of these expected requirements ensures the delivery of the package at the scheduled time. A drone provider request is defined as a tuple of $\langle SP_{id}, D_{id}, S, RQT \rangle$, where

- SP_{id} is a unique drone service provider identifier.
- D_{id} is the unique drone ID operating in the skyway network owned by SP_{id} .
- S is a set of skyway segments required by D_{id} to transit from one node to another. S is a tuple represented by $\langle seg_s, \dots, seg_d \rangle$, where each seg_i represents a skyway segment required by D_{id} . In each seg_i , D_{id} has expected

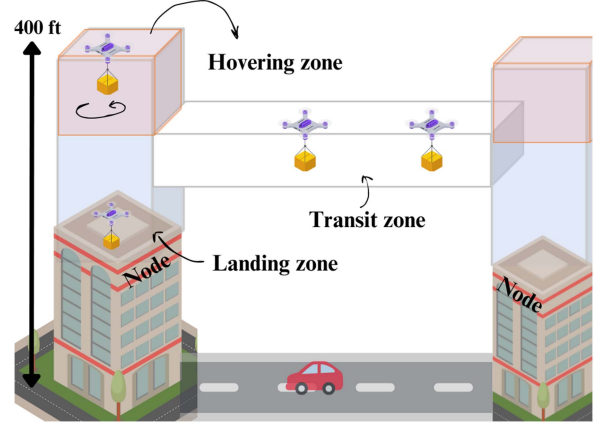


Fig. 3. Designated zones in a skyway network.

segment-level requirements represented as a tuple of $\langle EX(b), EX(T_{(x,y,z)}), EX(st) \rangle$, where:

- $EX(b)$ represents the expected battery consumption of a drone in the segment.
- $EX(T_{(x,y,z)})$ represents the expected trajectory of a drone while operating in seg_i . It is a sequence of waypoints $W_1, W_2, \dots, W_n$. Each waypoint is defined by a tuple of coordinates $\langle x_i, y_i, z_i \rangle$, where x_i , y_i , and z_i represent the latitude, longitude, and altitude of the waypoint, respectively.
- $EX(st)$ is the tuple of $\langle t_{start}, t_{end} \rangle$ where t_{start} and t_{end} are the expected start and end of the segment traversal times, respectively.
- RQT represents the time at which the service provider submits its request to the super-provider.

Definition 2 (Skyway Segment): The drones use skyway segment as a *transit zone* to operate between nodes (Fig. 3). We represent the attributes associated with the skyway segment structure in the form of a tuple as follows $\langle S_{id}, V, C_{max}, D_{threshold}, Alt_{range}, state_{seg} \rangle$, where

- S_{id} is a unique skyway segment ID.
- V is the volume of the segment, which is represented as a tuple of $\langle l, w, h \rangle$ where l, w, h is the skyway segment's length, width, and height, respectively.
- $D_{threshold}$ is the optimal distance between drones that allows them to operate efficiently in a segment.
- C_{max} is the maximum number of drones that may operate in the skyway segment without interference. It is computed as follows:

$$C_{max} = \frac{V}{V_d + V_{sep}} \quad (1)$$

where V_d is the volume of drones operating in Segment S_{id} . $V_{sep} = D_{threshold}^3$ is the separation volume required for each drone in 3D segment.

- Alt_{range} is the tuple of $\langle Alt_{max}, Alt_{min} \rangle$ where, Alt_{max} and Alt_{min} are the maximum and minimum altitudes, respectively at which drones may operate in a segment. The maximum altitude is usually defined by the

aviation authorities.³ The minimum altitude is dependent on the height of the rooftop building.

Definition 3 (Skyway segment state ($state_{seg}$)): The state of a skyway segment reflects the drone traffic condition in a segment at time ΔT . We identify different attributes of a segment that inform the state of a segment. We formally define the segment's state ($state_{seg}, \Delta T$) based on these attributes as a tuple of $\langle S_{id}, \Delta T, D_{sep}, D_{numb}, Cong_{level} \rangle$, where:

- S_{id} is the unique segment identifier.
- ΔT indicates the time interval in which the state of the skyway segment is monitored.
- D_{numb} is the total number of drones operating in a skyway segment S_{id} in time interval ΔT .
- $Cong_{level}$ indicates the level of congestion in a segment. We categorize the levels of congestion according to the Utilization Factor (UF) of a segment as follows:

$$Cong_{level}(t) = \begin{cases} \text{Free} & \text{if } UF = 0 \\ \text{Low} & \text{if } 0 \leq UF < 0.4 \\ \text{Medium} & \text{if } 0.4 \leq UF < 0.7 \\ \text{High} & \text{if } 0.7 \leq UF < 1 \\ \text{Critical} & \text{if } UF > 1 \end{cases} \quad (2)$$

where, UF is the ratio of the number of drones ($D_{numb,t}$) to the optimal capacity (C_{max}) of a segment. We calculate UD of a segment at any time t as follows:

$$UF(t) = \frac{D_{(numb,t)}}{C_{max}} \quad (3)$$

- $D_{sep}(t)$ indicates the spatial separation between drone pairs operating within a 3D airspace in a skyway segment at time t . We measure the lateral, longitudinal, and vertical separation between drone pairs based on their configuration using the following formulas:

– **Longitudinal separation:** The longitudinal separation between drone pair i, j operating at the same altitudes can be calculated using the following formula:

$$Long_{ij}(t) = |lon_j - lon_i| \quad (alt_i = alt_j) \quad (4)$$

where, $Long_{ij}$, is the longitudinal separation between drone pair i and j operating at the same altitude alt_i and alt_j with longitudes lon_i and lon_j respectively Fig. 4(a).

– **Lateral separation:** The lateral separation is the distance between drone pairs operating parallel to each other at the same altitude in the same skyway segment. The lateral separation Lat_{ij} between drone pair pair i, j can be computed using the following formula:

$$Lat_{ij}(t) = |lat_j - lat_i| \quad (alt_i = alt_j) \quad (5)$$

where, lat_i , and lat_j , is the latitude of drone i and drone j , respectively Fig. 4(a).

– **Vertical separation:** The vertical separation between drone pairs operating at different altitudes in parallel to each other in the same skyway segment can be calculated as follows:

$$Vert_{ij}(t) = |alt_j - alt_i| \quad (alt_j > alt_i) \quad (6)$$

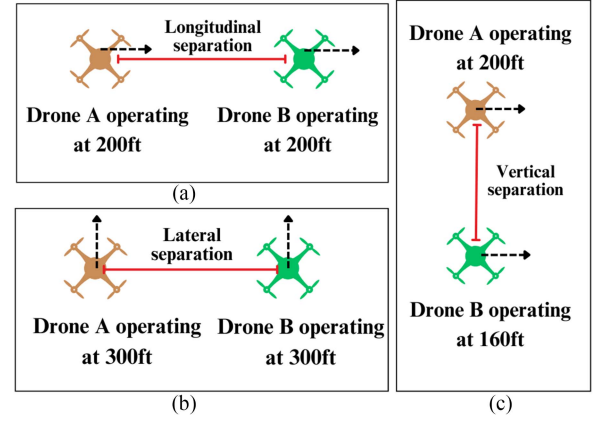


Fig. 4. Different types of drone configurations in a skyway segment (a) Front and rear (b) Side-by-side (c) Top-down.

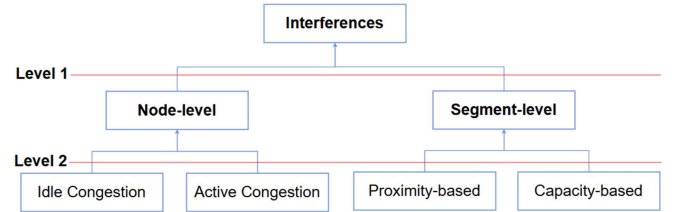


Fig. 5. Drone-service interference classification.

where $Vert_{ij}$ is the vertical separation between drones i and j operating at altitudes alt_i and alt_j , respectively Fig. 4(c).

– **Offset separation:** The offset separation between drone pairs is calculated using the following formula:

$$Xoffset_{ij}(t) = |lat_j - lat_i|$$

$$Yoffset_{ij}(t) = |lon_j - lon_i| \quad (7)$$

where $Xoffset_{ij}$, $Yoffset_{ij}$ is the offset between drone pair i, j along x and y axis, respectively.

A. Drone Services Interference Taxonomy

In the context of drone operations, an *interference* refers to the disruption in the transmission and reception of data between drones and control systems. However, we are defining *interference* as an event that hinders the *efficient* operation of drones in a skyway network. This hindrance affects the fulfillment of expected QoS requirements for drone services. We present a novel drone service interference taxonomy (Fig. 5). We categorize the interference into two main levels based on the location of occurrence. The first level specifies the location (i.e., node or skyway segment) of interference within the skyway network. The second level outlines different types of interference at both the node and segment levels. The *node-level interference* is out of the scope of this paper and we are mentioning it here as a part of the proposed taxonomy. Below we describe each type in detail:

³cas.gov.au/drones/drone-rules/emergencies-and-public-spaces

- **Node-level Interference:** This interference refers to any event that *delays* the provisioning of drones' recharging requirements at a node resulting in late package delivery. We assume that this delay manifests in the form of *congestion* at the node-level. We formally define a node-level interference (*NLI*) as follows:

$$NLI = \begin{cases} 1 & \text{if } \frac{\lambda}{R_{numb} \cdot \mu} > 1 \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

where λ and μ are the arrival rate and service rate⁴ of drones in the queue at node N_{id} with R_{numb} of recharging pads. We identify two types of congestion at the node-level.

- **Idle Congestion:** This interference is defined as a delay that impacts the expected arrival time of a drone at the subsequent node. It typically occurs when drones are idle at the waiting pads, waiting for their turn to occupy the recharging pad without consuming additional battery.

$$IC = \begin{cases} 1 & \text{if } \frac{\sum_{i=1}^n (W_i)}{n} > EX(NT_i) \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

where, W_i is the the waiting time and $[EX(NT_i)]$ denotes the expected arrival time of drone i at the subsequent node.

- **Active Congestion:** This interference refers to a delay that impacts both the expected battery requirement and the expected arrival time of the drone at the subsequent node. This interference usually occurs when drones have to hover in the hovering zone due to the unavailability of a waiting or recharging pad in the landing zone. Active congestion (*AC*) is defined as:

$$AC = \begin{cases} 1 & \text{if } (\delta[EX(b_i)] + \delta[EX(NT_i)]) > 0 \\ 0 & \text{otherwise} \end{cases} \quad (10)$$

where $\delta[EX(b_i)]$ and $\delta[EX(NT_i)]$ are the increase in battery consumption and delay in arrival time of the drone i at the subsequent node, respectively, due to hovering.

- **Segment-level Interference:** This interference refers to any event that impacts the optimal requirements of drones in a skyway segment. These requirements include the parameters (P), i.e., segment travel time (st), spatial coordinates $T_{(x,y,z)}$, and anticipated battery consumption level (b) in a skyway segment. The impact on these parameters (P) may result in *safety* or *efficiency* issues. We formally define segment-level interference (*SLI*) in terms of these parameters as follows:

$$SLI = \begin{cases} 1 & \text{if } \sum_{i=1}^n \delta[EX(P_i)] > 0 \\ 0 & \text{otherwise} \end{cases} \quad (11)$$

where, $\delta[EX(P_i)]$ indicates deviation from the expected parameter values of drone i in a skyway segment.

The segment-level interference is categorized as follows:

⁴The rate at which drones access and utilize the recharging pads at a node.

TABLE I
SUMMARY OF NOTATIONS

Symbol	Description
S_{id}	Unique skyway segment ID
t_{ij}^{overlap}	Temporal overlap between drones d_i and d_j (minutes)
C_{max}	Optimal capacity of a segment
D_{sep}	Separation distance between drone pair along 3D axis (meters)
$D_{threshold}$	Minimum separation along 3D required to avoid interference
SP_{ij}	Spatial proximity between drone pair i and j (meters)
SC_{ij}	Spatial configuration between drone pair ij (e.g., top-down)
EX	Expected values of a parameter (i.e., delivery time (DT), battery consumption in each segment (b_i), segment travel time (st)) anticipated by the drone provider without considering interferences
AC	Actual values of parameters (i.e., DT , b_i , st) computed by super-provider considering inter-drone interference

- **Capacity-based:** This interference occurs when the number of drones operating in a skyway segment is greater than its optimal capacity. Capacity-based Interference (*CI*) is defined as:

$$CI = \begin{cases} 1 & \text{if } f_{S,\Delta t} > C_{max}, \Delta t = 1, 2, \dots, N \\ 0 & \text{otherwise} \end{cases} \quad (12)$$

where, $f_{S,\Delta t}$ represents the observed flow of drone traffic during time interval Δt .

- **Proximity-based:** This interference occurs when the drones operate in close proximity such that their aerodynamic forces interfere with each other. This may negatively impact the anticipated battery consumption of drones. We define proximity-based segment-level Interference (*PI*) as:

$$PI = \begin{cases} 1 & \text{if } \exists \text{ pair } (i, j) \text{ where } (0 < D_{sep} < D_{threshold}) \\ 0 & \text{otherwise} \end{cases} \quad (13)$$

where D_{sep} represents the spatial separation between drone pair (i, j) in a 3D space.

In this paper, we will focus on the *detection* and *severity assessment* of proximity-based segment-level inter-drone interferences occurring in a skyway network. Table I provides a reference for the main symbols used in this paper.

Definition 4 (Interference Detection and Severity Assessment Problem): Given a set of n drone provider requests, our work aims to monitor the drone-based usage of skyway segments for the detection and severity assessment of inter-drone interference. This work aims to find the function F in $I, ISL \approx F(state_{seg}, DPR)$ where ISL is the corresponding severity level of an interference I .

V. DRONE SERVICE INTERFERENCE DETECTION FRAMEWORK

In this section, we present a drone service interference detection framework. Fig. 6 presents an overview of the framework. The proposed inter-drone interference framework consists of two main modules: (1) *Inter-drone Interference Detection* (2)

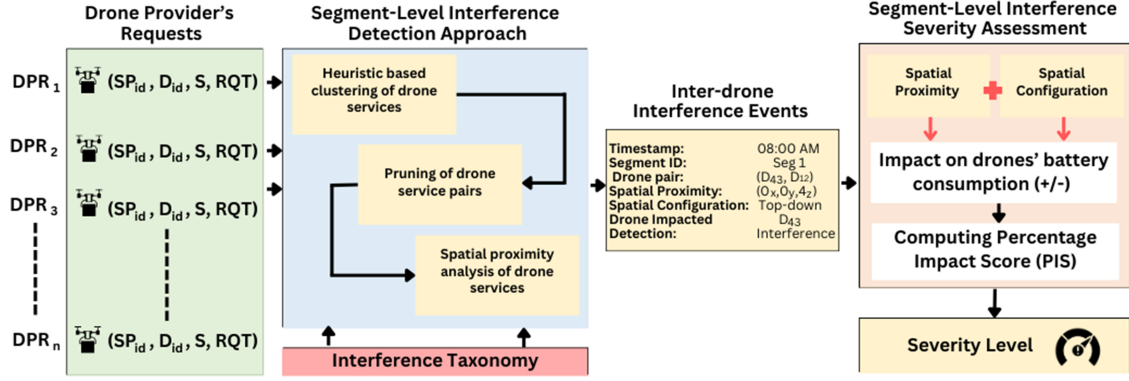


Fig. 6. Segment-level interference detection framework.

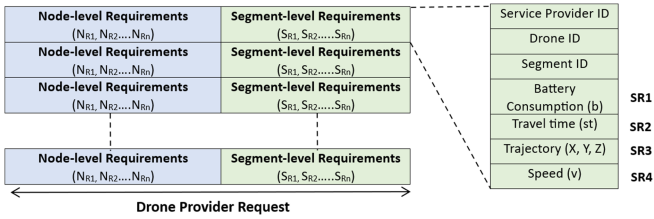


Fig. 7. Drone providers' request consisting of segment-level requirements.

Interference Severity Assessment (Fig. 6). In the first module, we develop an algorithm to analyze the spatial proximity between peer-to-peer drone combinations with spatio-temporal overlaps. The details of the proposed segment-level interference detection approach are illustrated in Algorithm 1. We use multiple heuristics to prune the solution space for evaluating drone pairs most likely to experience interference. In the second module, we assess the overall severity of interference events occurring in the segment.

A. Segment-Level Inter-Drone Interference Detection

A. Phase 1: Heuristic-based Clustering of Drone Services

The goal of this phase is to reduce the solution space for evaluating drone pairs for interference. The input of this phase is the set of drone providers' requests (DPR). These requests are the drone providers' optimal *requirements* to ensure on-time package delivery. Each drone provider submits segment-level requirements ($SLRs$) for n number of segments that the drone d would traverse. These requirements consist of the service provider ID (SP_{id}), drone ID (D_{id}), trajectory ($T_{(X,Y,Z)}$), segment travel time (st) and battery consumption (b) (Fig. 7). For example, a drone may require to operate within segment S , maintaining an altitude of 150ft and a speed of 45 km/h. As per the QoS requirement, the drone may expect to consume approximately 0.231 kWh of battery power during its segment travel time from 1:00 PM to 1:10 PM to arrive at the next node. Each drone service has varying segment-level requirements. We assume each drone provider submits a request ahead of time. In this phase, the algorithm clusters drones based on a heuristic

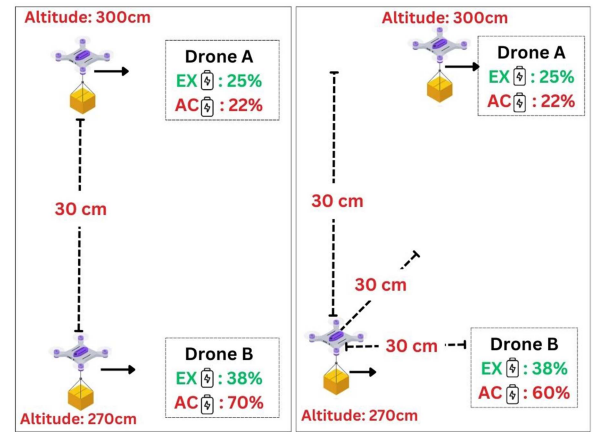


Fig. 8. Impact on the battery consumption of drones based on their spatial distribution (a) Drone A and B in top-down configuration (b) Drone A and B in top-down configuration with an offset of 30 cm .

(Lines 2-8). This heuristic is based on the intuitive notion that the drones scheduled to take off for the same destination nodes and in the same hourly interval ($\Delta t = 60$ minutes) are more likely to share the same path and interfere. This heuristic defines the solution space consisting of drone pairs most likely to experience inter-drone interference. We define this heuristic in (14).

$$C_{ij} = \begin{cases} 1 & \text{if } |t_i - t_j| \leq \Delta t \text{ and } D_i = D_j \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

where, t_i, t_j are the take-off times of drone i and drone j for their destination nodes D_i and D_j , respectively. Hence, the above heuristic will not group the drones in the same cluster if either their takeoff time does not overlap within the time interval Δt or the destination node is different. It is then followed by the formation of peer-to-peer drone pairs within the same cluster (Lines 9-11). The drone pairs within the same cluster are expressed as:

$$P(C_i) = \{(d_p, d_q) \mid d_p, d_q \in C_i, p \neq q, \text{ for } i = 1, 2, \dots, k\} \quad (15)$$

Algorithm 1: Detection of Segment-Level Inter-drone Interference.

Require: *DPR* (Drone providers' requests with segment-level requirements)
Ensure: *IDS* (Set of drone services experiencing interference),
CDS (Set of drone services experiencing collision)

```

1: Clusters =  $\emptyset$ , CandidateDronePairs =  $\emptyset$ ,  

   IDS =  $\emptyset$ , CDS =  $\emptyset$ ,  $\Delta t = 60$ 
{Phase 1: Heuristic-Based Clustering of Drone Services}
2: for each  $d_i$  in DPR do
3:   for each  $d_j$  in DPR do
4:     if  $|t_i - t_j| \leq \Delta t$  end  $D_i == D_j$  then
5:       Clusters  $\leftarrow$  Clusters  $\cup \{d_i, d_j\}$ 
6:     end if
7:   end for
8: end for
9: for each cluster  $C_k$  in Clusters do
10:   $P(C_k) \leftarrow \{(d_p, d_q) \mid d_p, d_q \in C_k, p \neq q\}$ 
11: end for
{Phase 2: Pruning Drone Services Based on Spatio-Temporal Overlap}
12: for each  $(d_p, d_q)$  in  $P(C)$  do
13:   if  $S_p = S_q$  end  $[t_{p,start}, t_{p,end}] \cap [t_{q,start}, t_{q,end}] \neq \emptyset$  then
14:     CandidateDronePairs  $\leftarrow$   

       CandidateDronePairs  $\cup \{(d_p, d_q)\}$ 
15:   end if
16: end for
{Phase 3: Spatial Proximity Analysis Between Drone Pairs}
17: for each dronePair  $(d_p, d_q)$  in CandidateDronePairs do
18:    $Long_{pq}(t) \leftarrow |x_q - x_p|$  if  $z_p = z_q$  (4)
19:    $Lat_{pq}(t) \leftarrow |y_q - y_p|$  if  $z_p = z_q$  (5)
20:    $Vert_{pq}(t) \leftarrow |z_q - z_p|$  if  $z_q > z_p$  (6)
21:    $X_{offset_{pq}}(t) \leftarrow |y_q - y_p|$ 
22:    $Y_{offset_{pq}}(t) \leftarrow |x_q - x_p|$  (7)
23:   if  $(Lat_{pq} == 0)$  and  $(Vert_{pq} == 0)$   

     and  $(Long_{pq} == 0)$  and  $(X_{offset_{pq}} == 0)$  and  

      $(Y_{offset_{pq}} == 0)$  then
24:     CDS  $\leftarrow$  CDS  $\cup \{(d_p, d_q)\}$ 
25:   else if  $(0 \leq Long_{pq} \leq D_{threshold})$  and  

      $(0 \leq Lat_{pq} \leq D_{threshold})$   

     and  $(0 \leq Vert_{pq} \leq D_{threshold})$  and  

      $(0 \leq X_{offset_{pq}} \leq D_{threshold})$  and  

      $(0 \leq Y_{offset_{pq}} \leq D_{threshold})$  then
26:     IDS  $\leftarrow$  IDS  $\cup \{(d_p, d_q)\}$ 
27:   end if
28: end for
29: return IDS, CDS

```

In this equation, $P(C_i)$ represents the set of drone pairs within cluster C_i . Each pair (d_p, d_q) in P consists of drones d_p and d_q belonging to cluster C_i . Here, the condition $p \neq q$ ensures that each pair is unique. The output of this phase will be a set of heuristic-based drone pairs that are likely to experience interference in segments.

A. Phase 2: Drone Services' Pruning based on Spatio-temporal Overlapping

The goal of this phase is to prune drone services based on spatio-temporal overlapping. The input of this phase is a set of drone service pairs from the previous phase (15). In this phase, the algorithm aims to prune the drone services from each cluster that does not reflect spatio-temporal overlapping (Lines 12-16). The spatio-temporal overlapping between two drone services occurs when the drones share the *same segment* within the *same time interval* (Line 13). For instance, $(d_1, d_2, S_3, 10:00 \text{ AM} - 10:10 \text{ AM})$ shows that d_1 and d_2 share segment S_3 for the time interval between 10:00 AM to 10:10 AM. Given two drones d_p and d_q operating in segments S_p and S_q during time intervals t_p and t_q , the spatio-temporal overlap between this drone pair is defined as follows:

$$[b]P(d_p, d_q, S, t) = \{(d_p, d_q) \mid S_p = S_q \wedge [t_{p,start}, t_{p,end}] \cap [t_{q,start}, t_{q,end}] \neq \emptyset\} \quad (16)$$

A. Phase 3: Spatial Proximity Analysis between Drone Pairs

The goal of this phase is to detect proximity-based interference between drone pairs. The input of this phase is the set of drone pairs from the previous phase that share the same segment within the same time interval. However, spatio-temporal overlap alone does not necessarily indicate interference, as drones may operate at different coordinates within the same segment and time. For each drone pair, we compute the separation between the drones along x, y, z axis using (5)–(7) (Lines 17-22). Moreover, there may be instances of collision where a drone pair shares the same coordinates simultaneously, resulting in zero separation along the x, y, z axes (Lines 23- 24). Hence, a proximity-based interference event occurs if the spatial separation D_{sep} between drone pair d_i and d_j along x, y, z axis is greater than 0 but less than a certain separation threshold ($D_{threshold}$) (refer to (13)) (Lines 25-26). These thresholds vary along the 3D axes based on the spatial configuration of the drones (e.g., top-down, front-rear, side-by-side) within the segment. The output of this phase is the set of all drone services experiencing interference *IDS* and collision *CDS* in the time interval Δt in segment S (Line 29).

B. Segment-Level Interference Severity Assessment

The input of this module is the set of all drones experiencing proximity-based interference from the peer drones. This module assesses the overall severity level based on the cumulative impact of inter-drone interference events occurring in a segment. To quantify the overall severity level of interference, we first compute the impact of interference on each drone relative to its peer drones operating in the same segment. In this regard, we compute the impact of interference on the drones' *battery consumption*. We then *assess* how this updated battery consumption due to interference affects the *expected delivery time* of the drone. We use (17) from the model proposed in [24] to compute the expected battery consumption ($EX(b_i), S_k$) of the drone d_i as

it traverses segment S_k .

$$EX(b_i, S_k) = \frac{d_i^{S_k}}{V_{d_i}^{S_k}} \left(\frac{(M_{d_i} + m_{d_i}) \times V_{d_i}^{S_k}}{\frac{3600}{g}(\eta r)} + pw \right) \quad (17)$$

where, $d_i^{S_k}$ is the distance traversed by the drone in the skyway segment, $V_{d_i}^{S_k}$ is the speed of the drone, M_{d_i} represents the mass of the drone, m_{d_i} is the payload mass being carried. The gravitational acceleration constant is denoted by g , η refers to the motor/propeller power transfer efficiency, r represents the lift-to-drag ratio of the drone and pw is the power consumption of the drone's electronics. The *impact on battery consumption* is determined based on the percentage change in expected battery consumption ($EX[b]$) of the drone i due to interference from the peer drone using (18).

$$\delta(EX[b_i]) = \frac{AC(b_i) - EX(b_i)}{EX(b_i)} \quad (18)$$

where, ($AC(b_i)$) is the actual battery consumption of the drone i due to interference from n number of peer drones operating in segment S_k for time interval $t_{overlap}$. It can be computed using the formula given in (19):

$$AC(b_i, S_k) = EX(b_i, S_k) + \sum_{j=1}^n (I_{ij} \times t_{ij}^{overlap}) \quad (19)$$

where, I_{ij} is the additional battery drain due to inference from the peer drone j . We identify two main factors that cumulatively inform battery drain due to proximity-based interference on the drones' expected battery consumption: (1) Spatial Proximity (SP) (2) Spatial Configuration (SC). We compute I_{ij} using the following formula based on these factors:

$$I_{ij} = f(SP_{ij}, SC_{ij}) = \begin{cases} \alpha_1 & \text{if } SP_{ij} \leq s_1 \\ \alpha_2 & \text{if } s_1 < SP_{ij} \leq s_2 \\ \alpha_3 & \text{if } s_2 < SP_{ij} \leq s_3 \\ 0 & \text{if } SP_{ij} > s_3 \end{cases} \quad (20)$$

where, $\alpha_1, \alpha_2, \alpha_3$ is the additional drain in the battery for different separation ranges between drone pair ij . We formulate equations given in [25] for all possible spatial configurations and proximities between drone pairs. For each configuration and proximity, we compute the values of $\alpha_1, \alpha_2, \alpha_3$ using the dataset collected in [9]. The first factor (i.e., SP_{ij}) refers to the relative separation between drones i and j along the x, y , and z axes in the 3D segment. The second factor (i.e., SC_{ij}) refers to how drones i and j are positioned relative to each other within a segment S_k . The drones can have random spatial configurations at time t i.e., top-down, side-by-side, front-rear, or offset formations. We use (4)–(7) to calculate the spatial proximity between drone pairs based on their configuration with the peer drone. For instance, consider two drones i and j sharing the same segment SEG 100 for an interval of $t_{overlap} = 300$ seconds. These drones operate in parallel trajectories along the x - axis with a vertical separation of 30cm along the y -axis. It indicates that $SC_{ij} = \text{Top-Down Configuration}$ with drone i operating directly above drone j with $SP_{ij} = 30$ cm. Hence, the impact of interference due to drone j on the drone i battery consumption would be $AC(b_i) = 22\% + (0.18 * 300) = 54\%$.

Therefore, the impact of interference on drone i would be $\delta(EX[b_i]) = \frac{54-22}{22} = 145\%$ ((18)). It indicates that drone i consumes an additional battery of 32% due to interference from peer drone j operating in close spatio-temporal proximity. As a result, the drone may have to spend more time at the intermediate node to regain the drained battery due to interference from the peer drone. Consequently, it impacts the expected delivery time of the drone. We find the *impact on the expected delivery time* of drone using the following equation:

$$\delta(EX[DT_i]) = \frac{AC(DT_i) - EX(DT_i)}{EX(DT_i)} \quad (21)$$

where, ($EX(DT_i)$) and $AC(DT_i)$ is the expected and actual delivery time of the drone i . The *expected delivery time* is estimated by the drone provider based on the intrinsic and extrinsic constraints associated with the environment. The *actual delivery time* is based on the increase in recharging time of the drone at the n number of intermediate nodes due to the occurrence of proximity-based interference in the segments. We compute the actual delivery time as follows:

$$AC(DT_i) = EX(DT_i) + \sum_{k=1}^n (RT_{ki}) \quad (22)$$

where, RT is the updated charging time required by the drone to reach from its initial battery state (i.e., $AC(b_i, S_k)$) to final battery state (i.e., S_f). It can be calculated using (23).

$$RT = -\tau \cdot \ln \left(\frac{1 - S_f}{1 - AC(b_i, S_k)} \right) \quad (23)$$

where $\tau \approx T_{full} * \beta$ is the function defined in terms of the time required by the drone to recharge fully i.e., $S_f = 99\%$. Assume a drone has an actual battery consumption of 54% due to interference and it requires 40 minutes to recharge fully then $\tau = 40 * 0.2 = 8$ and $RT = 8 \cdot \ln \frac{1-0.99}{1-0.54} \approx 30$ minutes.

In the next step, we quantify the overall severity of interference events occurring in a segment S_k at time t . The degree of severity due to multiple inter-drone interferences in a segment S_k is measured on a scale from 0 to 100, which is categorized into three levels; low, medium, and high. These levels reflect the severity of an interference in a skyway segment. The identification of the interference severity level allows the determination of appropriate interference resolution strategies. We determine the severity level of inter-drone interferences occurring in a segment based on the Average Percentage Impact Score (PIS). Average Percentage Impact Score indicates the average impact on the expected battery consumption ($EX(b)$) of drones experiencing interference. The categorization is based on the idea that the threshold of 40% indicates a higher risk to drone operations from the super provider's perspective. We categorize the severity levels of interference in a segment S_k at any time t as follows:

$$ISL(S_k, t) = \begin{cases} \text{Low,} & \text{if, } (0 < PIS \leq 20\%) \\ \text{Medium,} & \text{if, } (20\% < PIS \leq 40\%) \\ \text{High,} & \text{if, } (PIS \geq 40\%) \end{cases} \quad (24)$$

We compute the average Percentage Impact Score ($PIS(S_k, t)$) of segment S_k with n number of drones experiencing interference at time t as follows:

$$PIS(S_k, t) = \frac{\sum_{i=1}^n (\delta(EX[b_i]))}{n} \quad (25)$$

where, $\delta(EX(b_i))$ is the deviation from the expected value of drone i battery consumption ($EX[b_i]$) due to interference. Consider two drones $d1$ and $d2$ operating in close proximity with the same expected battery consumption values i.e., $EX(b_1) = EX(b_2) = 30$ Wh. Assume their actual battery consumption deviates from 30 Wh to 90 Wh due to inter-drone interference. We use (18) to compute the impact on the expected battery consumption of drones due to interference. This interference results in both drones consuming 200% more battery in actual. Hence, the overall severity level of interference is categorized as *high* with the PIS calculated as $(200 + 200)/2 = 200\%$ using (24).

VI. EXPERIMENTATION AND RESULTS

In this section, we present the details of our dataset and discuss the results of each phase from our proposed framework. We then evaluate the performance of our proposed interference detection approach in terms of accuracy, precision, recall, and execution time. We evaluate our approach against two approaches: baseline exhaustive and K-means. We ran all the experiments using Python on a DELL system equipped with a 12th Gen Intel(R) Core(TM) i7-1255 U processor (1.7 GHz, 7 cores) and 16 GB of memory.

A. Dataset Description

1) *Drone Delivery Dataset*: In our experiments, we use a real-world skyway network dataset based on the 3D model of Sydney CBD [26]. This network consists of 30 nodes and 75 LoS-based skyway segments. The skyway segment serves as a delimited airspace for drones to transit between nodes. The dimensions of each segment are based on the compact size of the Crazyflie drones (i.e., $92 \times 92 \times 29$ mm) and the constraints of operating within an enclosed indoor environment due to government restrictions on outdoor drone testing. The length of each segment corresponds to the distance between the nodes, and the height is uniformly set to 100 cm to align with the vertical clearance of the testbed. The width of each segment ranges from a minimum of 10 cm to a maximum of 100 cm. The former is defined based on the width of the Crazyflie drone (i.e., 9.2 cm) for safe traversal of drones through the segment airspace. The latter is defined to prevent overlap with the drone traffic of nearby segments, which may lead to collisions or near-misses.

We use Crazyflie 2.1⁵ as the drones operating in the skyway network due to their compact size and suitability for repetitive indoor experiments. The trajectory of each drone from the source to the destination node is based on A*(A-star) algorithm to ensure the shortest path consisting of nodes and segments. Each

TABLE II
EXPERIMENTAL VARIABLES

Variable	Value
Drone model	Crazyflie 2.1
Drone dimensions	92 x 92 x 29 mm
Mass of the drone	27 g
Maximum payload capacity	15 g
Maximum drone flight time	≈ 7 min
Maximum drone speed	1 m/s
Recharging time from 0% to 100%	≈ 40 min
Indoor testbed dimensions	4690 x 3860 x 2700 mm
Maximum number of requests	2000
Drone flight duration (min)	6-74 (Avg: 33)
Drone flight distance (mm)	760-5520 (Avg: 2700)
Number of nodes visited per flight	3-8 (Avg: 4)

drones' trajectory in a segment is a sequence of waypoints $W_1, W_2, \dots, W_n$, where each W_i is defined by a tuple of 3D coordinates $\langle (x_i, y_i, z_i) \rangle$, that the drone traverses at a specific timestamp. We compute the energy consumption for each drone flight using (17) based on the battery consumption model proposed in [24]. The recharging duration at the intermediate nodes is computed using (23) if the drone's battery is insufficient to traverse the next segment. We use order placement times from the food delivery dataset of Bogotá City [27] as the drone flight start times. We then specify a set of 400 to 2000 drone flights considering different payloads, flight ranges, and flight start times. The experimental variables are described in Table II

2) *Inter-Drone Interference Dataset*: We performed experiments in an indoor drone testbed to find the impact on drones' battery power consumption as it operates concurrently with the peer drone between nodes. The experiments were conducted for varying spatial positions and proximities between two drones as they traversed the segment. In this regard, we first collected data on the battery power consumption of two drones when they operated individually (i.e., *solo*) in a segment. Afterward, we collected data on the battery power consumption of both drones as they operate concurrently (i.e., *simultaneously*) in the same segment. We then compute the change in battery power consumption of the drones for both *solo* and *simultaneous* flights between nodes. This change in battery consumption serves as the impact of inter-drone interference on the expected battery consumption of the drone. For the inter-drone interference dataset, we collected the impact on drones' battery power consumption for varying spatial configurations, i.e., top-down, front-rear, side-by-side, top-down offset, and side-by-side offset between drones in a 3D segment. We also conducted the experiments with varying inter-drone distances, i.e., 30 cm, 40 cm, 50 cm, 80 cm, and 100 cm, for each of the aforementioned spatial configurations. The experimental setup and dataset details are available at [9] and [25], respectively. We then use the results from the collected dataset to compute the impact of inter-drone interference on each drone flight in the drone delivery dataset.

3) *Relatability of Indoor Drone Testbed to Outdoor Settings*: We setup an indoor drone testbed due to government restrictions on conducting outdoor experiments with commercial drones [28]. While the testbed replicates a scaled-down 3D model of Sydney's CBD, with the compact size Crazyflie

⁵<https://www.bitcraze.io/products/crazyflie-2-1/>

as the drones operating in the network, it closely mimics the real-world settings [26]. The small-sized Crazyflie drones share the same multi-rotor architecture as the large-sized drones (e.g., DJI Flycart 30, Amazon MK30) used by companies like Amazon and DHL for package delivery [29]. Therefore, they exhibit similar aerodynamic behavior, which includes the generation of turbulent airflow due to continuous propeller rotation, which may impact the performance of nearby drones operating in close proximity [9]. However, the magnitude and spatial proximity at which drones may impact would scale up due to the magnitude of turbulence, which increases with the increase in the size of propellers [30]. Thus, the interference behavior observed in the testbed is reliable, as it captures the aerodynamic dynamics that are expected to scale with drone size in practical deployments.

Although the segment dimensions are modeled based on the small-size Crazyflie drones operating in the indoor settings, they would scale up given the large-size drones with a wingspan of 1980 mm expected to operate in urban skyway networks [31]. For instance, the length of the segment would correspond to the actual distance between the building rooftops (i.e., nodes), and the height would be defined based on the aviation authority regulations.⁶ The segment width would be proportional to the minimum of the width of the building rooftops, which serve as the endpoints of the segment. This would ensure that drones do not collide with the drones operating in the neighboring segments upon segment traversals between nodes. Albeit the expansion of segment dimensions, the skyway segments remain a delimited airspace given the complexity of skyway network infrastructure and predicted volume of drone traffic operations [32], [33]. Therefore, the indoor testbed closely mimics the delimited nature of the skyway segments where drones would follow a LoS path between nodes.

B. Results

In this section, we will first discuss the experiments conducted to observe how an increase in inter-drone interference events impacts the average delivery time of drones operating in the network. Then, we will discuss the effectiveness and efficiency of our proposed approach.

We conduct an experiment to evaluate how an increase in the number of drone delivery flights (ranging from 400 to 2000) operating in the shared network impacts the occurrence of inter-drone interference and collision events. We use order placement times from the food delivery dataset of Bogotá City as the request times of the drone delivery flights [27]. We then map the total number of requests over a 6-hour time duration across three rush hour periods: morning (between 8 AM and 10 AM), noon (between 12 PM and 2 PM), and afternoon (between 6 PM and 8 PM). We then assess the number of drone delivery flights experiencing interference and collision as the number of drones operating in the network increases. As shown in Fig. 9, interference and collision events increase as the number of drones operating in the network increases. The results show that $\approx 45\%$ of drone trajectories experience inter-drone interference

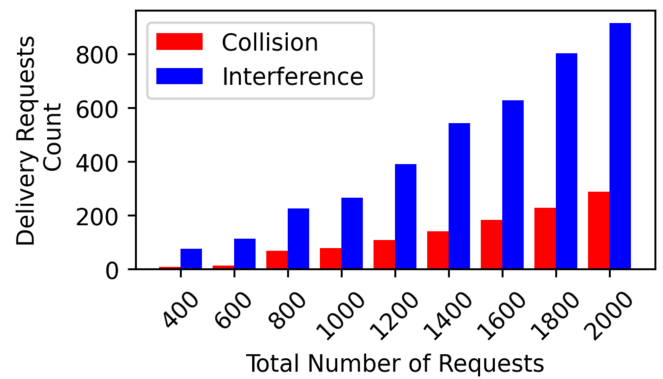


Fig. 9. Interference instances vs number of requests.

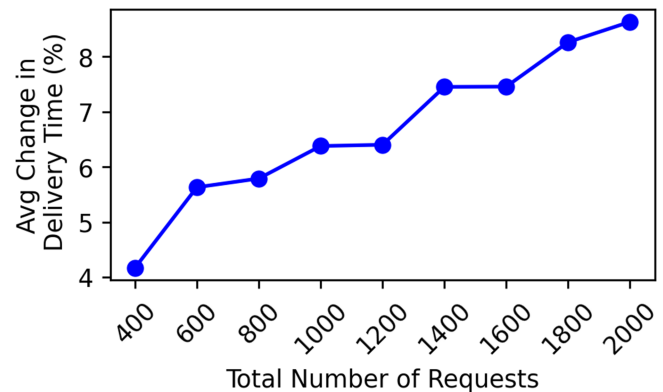


Fig. 10. Average percentage impact on delivery time.

in a network with 2000 drones. It is due to the increase in the spatio-temporal overlap between drones in a segment. Moreover, $\approx 15\%$ of the drones experience collisions due to a complete spatio-temporal overlap with other drones operating in shared skyway segments.

In the second experiment, we examine how these interference events impact the overall delivery time of drones operating in the network. In this regard, we compute the change in the expected delivery time of drones due to the occurrence of interference using (21). As shown in Fig. 10, the average change in the expected delivery time of drones increases with an increase in the number of drone trajectories operating in the network. It is due to increased congestion levels of drone traffic in a segment. As a consequence, drones operate in close spatio-temporal proximity, leading to increased energy consumption in a shared segment. This increased energy consumption necessitates more recharging time at intermediate nodes, thus leading to delays in package delivery.

We discuss an instance of the output from our proposed framework. Figs. 11 and 12 shows the state of two segments with IDs SEG77 and SEG220 for a time interval of 12:20 PM to 12:44 PM. The results show that the congestion level (11) in a segment does not reflect the severity level of an interference. For instance, a *medium* congestion level in a segment may not result in interference occurring between drones. This is because drones may not be operating in close proximity such that their battery

⁶casa.gov.au/drones/drone-rules/emergencies-and-public-spaces

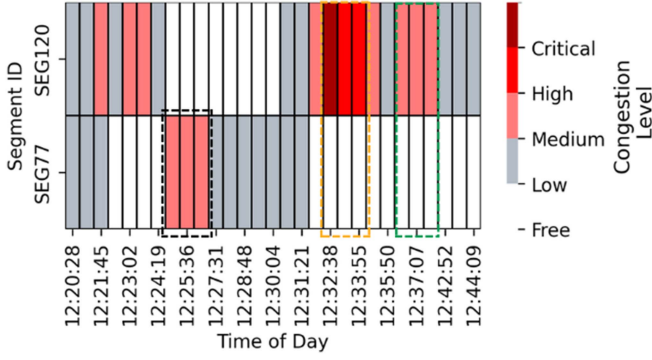


Fig. 11. Congestion level heatmap of segments (SEG77 and SEG120).

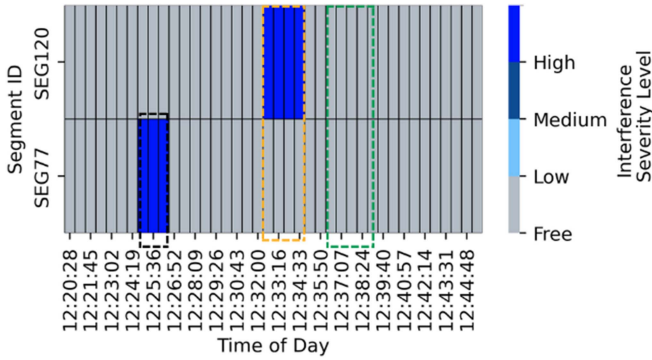


Fig. 12. Interference severity level heatmap of SEG77 and SEG120.

consumption gets impacted. As shown in Fig. 11, the congestion level of SEG120 between 12:36 PM and 12:38 PM is *medium*. However, the severity level of interference is *null* as the drones do not operate in close proximity (Fig. 12). Furthermore, the results reveal that the *critical* congestion level indicates that the segment is operating beyond its optimal capacity. As discussed in Section. IV-A, the maximum number of drones that may operate at an optimal distance without getting impacted by other drones determines the optimal capacity of the segment (refer to (1)). Fig. 11 illustrates a critical congestion level in SEG120 from 12:32 PM to 12:33 PM. As a result, the interference severity is high, as depicted in Fig. 12. This is because drones are operating in close spatial proximity such that their battery consumption is impacted critically.

C. Performance Evaluation

In this section, we evaluate the effectiveness and efficiency of our proposed approach in terms of accuracy, precision, recall, and computational time. We conducted experiments to compare the performance of our proposed heuristic-based interference detection approach against the baseline exhaustive and K-means clustering approach. Our baseline approach is an exhaustive approach that considers all possible drone pair combinations as a solution space to detect inter-drone interferences. We use the K-means clustering approach to perform spatio-temporal clustering of drone services based on the heuristics discussed in (14).

Effectiveness: We conduct a third experiment to evaluate the performance of our proposed approach considering the different numbers of drones operating in the network. The baseline approach exhaustively checks the spatio-temporal overlapping between all possible drone-pairs combinations. Therefore, we consider the interference detection results from the baseline exhaustive approach as ground truth values in these experiments. Fig. 13(a) and (c) show that our proposed time-based heuristic and K-means approaches have optimal *accuracy* and *recall* scores. In this context, *recall* measures the proportion of true interference events that the algorithm correctly detects out of all actual interference events. The results show that both approaches accurately identified all the inter-drone interference events in a segment (Fig. 13(a) and (c)). However, these approaches have a slightly low *precision* score of 88% in comparison to the baseline exhaustive approach (Fig. 13 (c)). *Precision* measures the proportion of correctly classified interference events out of all instances that the algorithm detected as interference. This *precision* score indicates that the proposed approach classifies some *non-interference* events as *interference events*. However, the proposed approach correctly detects all true inter-drone interference events (as evidenced by the 100% recall). It is crucial to detect all interference events accurately in scenarios where missing an inter-drone interference event could have a critical outcome with the cost of a few misclassifications. For instance, in scenarios where drones deliver medical packages, any undetected interference could delay essential deliveries, potentially risking lives.

Efficiency: In the fourth experiment, we evaluate the efficiency of our proposed approach in terms of computational time. We use the run-time efficiency metric to measure the algorithm's computational speed. The run-time efficiency (*RTE*) is calculated as the following:

$$RTE = \frac{1}{ET} \quad (26)$$

where *ET* (execution time) is the number of drone pair instances processed per unit time. Fig. 14(a) shows that the proposed time-based heuristic approach outperforms both the baseline exhaustive and K-means approaches in terms of execution time. This is because these approaches differ significantly in defining their solution space consisting of drone pairs for detecting interference events. Defining solution space is critical as it directly influences the *efficiency* of the approach. It is computationally expensive to consider all possible peer-to-peer drones for detecting interferences. Therefore, both the proposed approach and the K-means approach make a set of clusters based on the heuristic defined in (14), followed by the peer-to-peer drone pairs within the same cluster using (15). We use the elbow method to determine the optimal number of clusters for defining the solution space in the K-means approach [34], [35]. As a result, the proposed approach makes a greater number of clusters in comparison to the K-means approach, as shown in Fig. 14(c). This is because K-means optimizes the clusters based on minimizing intra-cluster variance using numerical data (i.e., takeoff times) while treating categorical values (i.e., destination nodes) less granularly. Therefore, the proposed approach results in fewer

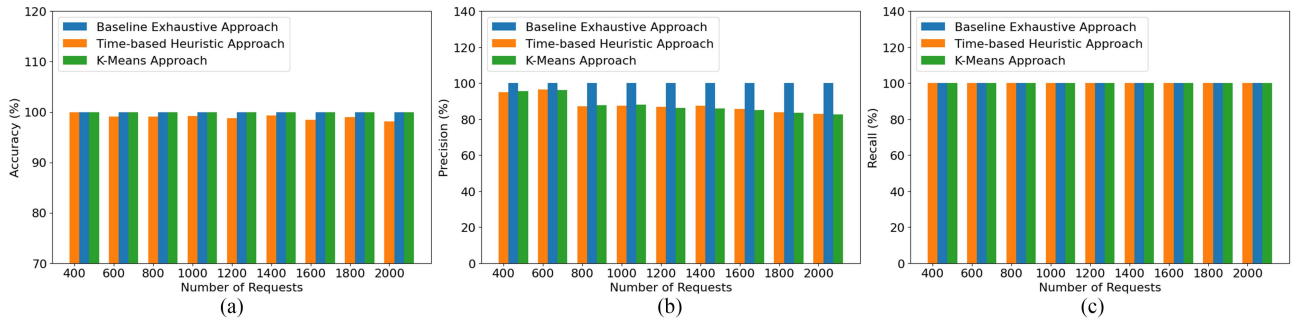


Fig. 13. Comparison between baseline exhaustive, K-means, and time-based heuristic approach in terms of accuracy (a), Precision (b), and Recall (c).

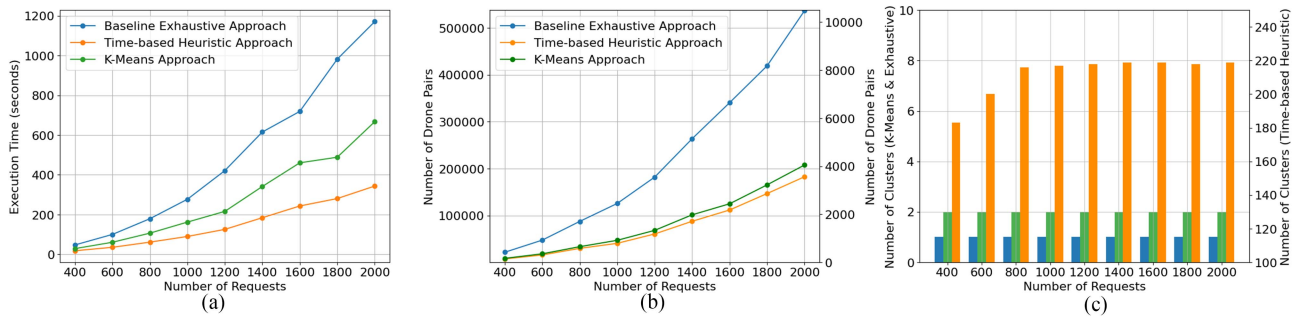


Fig. 14. (a) Execution time vs number of requests (b) Solution space vs number of requests (c) Optimal number of clusters vs number of requests.

peer-to-peer drone pairs for detecting interferences in comparison to K-means which leads to fewer but large-sized clusters as all the possible drone pair combinations are considered for each cluster (Fig. 14(b) and (c)). Therefore, the computational time required for spatio-temporal proximity analysis of drone pairs in each cluster is significantly reduced in the proposed approach compared to the K-means approach. This makes the proposed approach computationally efficient for time-sensitive transportation of medical supplies, where detecting interferences promptly ahead of time is crucial. Although both approaches perform equally effectively in detecting segment-level interference events. However, the proposed heuristic-based approach performs $\approx 70\%$ faster than the baseline exhaustive approach and $\approx 48\%$ faster than the K-means approach.

VII. CONCLUSION

We propose a novel inter-drone interference detection framework as a prerequisite to ensure the resilient operation of drone services. The proposed framework aims to detect and assess the impact of inter-drone interference on the expected QoS requirements of drone services. The proposed framework considers clustering-based spatio-temporal comparison between drone pairs in a 3D space to detect inter-drone interferences. In addition, the proposed approach assesses the impact of interference on the drones' energy consumption. The experimental results reveal that the proposed clustering-based approach for interference detection significantly reduces the computational time while providing optimal results. Our future work will

focus on resolving inter-drone interferences to maximize the satisfaction of drone providers using the skyway network.

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